Real records acquired before modeling

This report summarizes source profiling performed for Phase 1.

Five U.S. balancing authorities (CISO, ERCO, ISNE, NYIS and PJM), with publisher operating dates from 1 January 2024 through 31 December 2025. The acquired baseline contains 87,720 authority-hour records over 731 operating dates.

Official file	Original rows	Selected rows
EIA930_BALANCE_2024_Jan_Jun.csv	266,398	21,835
EIA930_BALANCE_2024_Jul_Dec.csv	269,426	22,085
EIA930_BALANCE_2025_Jan_Jun.csv	264,934	21,715
EIA930_BALANCE_2025_Jul_Dec.csv	267,218	22,085

Authority	Region	Distinct UTC hours	Operating dates
CISO	CAL	17,544	731
ERCO	TEX	17,544	731
ISNE	NE	17,544	731
NYIS	NY	17,544	731
PJM	MIDA	17,544	731

Acquisition route

Use the official bulk CSV route for the baseline. It supplies both business processes in one source record, keeps raw headers visible and requires no API key. The EIA reference workbook supplies names and region labels. The API is a possible later acquisition optimization after its units, timestamps and category crosswalk are tested against the bulk baseline.

The original four downloads contain 1,067,976 rows and 185,492,880 bytes. The included selected CSVs preserve all original fields for the five authorities and add two lineage fields. The manifest records complete-source and selected-subset SHA256 checksums. Original files can be downloaded again with the script.

Reference evidence

The official EIA930_Reference_Tables.xlsx is included. The authority_reference.csv extract reproduces its five selected BA rows. The nineteen-category energy mapping is derived from the actual headers across the four source files.

Source quality affects the design

Counts describe the frozen source extract. They are not PostgreSQL load results.

Check	Observed result	Design response
Identity	87,720 unique authority/UTC pairs; 0 duplicates.	Use authority + UTC instant as the demand business key.
Time coverage	0 missing UTC hours inside each authority’s acquired span; 2 repeated local clock labels per authority.	Keep the UTC key and local display fields.
Core blanks	218 blanks each for actual demand, forecast and total generation.	Keep nullable measures and report coverage.
Zero measures	5 actual zeros; 92 forecast zeros; 5 total-generation zeros.	Retain raw zeros. Forecast scores require actual > 0.
Numeric fuel cells	727,720 numeric cells available for generation facts; 31,007 are negative.	Keep signed net values and source classification.
Header sets	44 original columns in the first file; 65 in each later file. Fuel headers change from 9 to 16.	Map original header meanings explicitly; 19 distinct categories across files.
Reconciliation	87,471 hours comparable. 6,958 exceed max(10 MW, 1% of absolute reported total).	Flag for review; restrict matched-share metrics and disclose exclusions.

The reconciliation compares only numeric reported fuel cells with reported total generation. It does not substitute imputed or adjusted values. A nonzero residual may reflect reporting or category issues. The threshold is an explicitly chosen project diagnostic and is not proof that the source is wrong.

How to reproduce the evidence

Python 3.10 or later. Acquisition and profiling use only the Python standard library.

Action	Command or result
Download the baseline	python download_data.py --years 2024 2025
Profile the downloaded subset	python profile_data.py
Re-run using cached originals	Repeat the download command. Verified cache hashes avoid unnecessary network transfers.
Fetch a later half-year	python download_data.py --years 2026 --halves Jan_Jun --out new_extract
Recheck a revised file	Use --refresh and a new --out folder. Preserve the earlier folder and manifest.
Inspect evidence	Open data/sample_20_rows.csv, source_manifest.json and profile.json. Full selected raw CSVs are also included.

Source boundaries

The script filters authority rows and adds source locators. It preserves the source measures as text and does not correct values or load warehouse tables. The sample CSV uses a union of headers for viewing. Each complete selected file retains its own original schema, which remains authoritative for distinguishing absent categories.

The fixed 2024-2025 operating-date baseline includes some UTC end instants on 1 January 2026, because the source uses local operating days. Filter the baseline by Data Date when checking the two-year scope.

Public source roots
https://www.eia.gov/electricity/gridmonitor/sixMonthFiles/
https://www.eia.gov/electricity/930-content/EIA930_Reference_Tables.xlsx